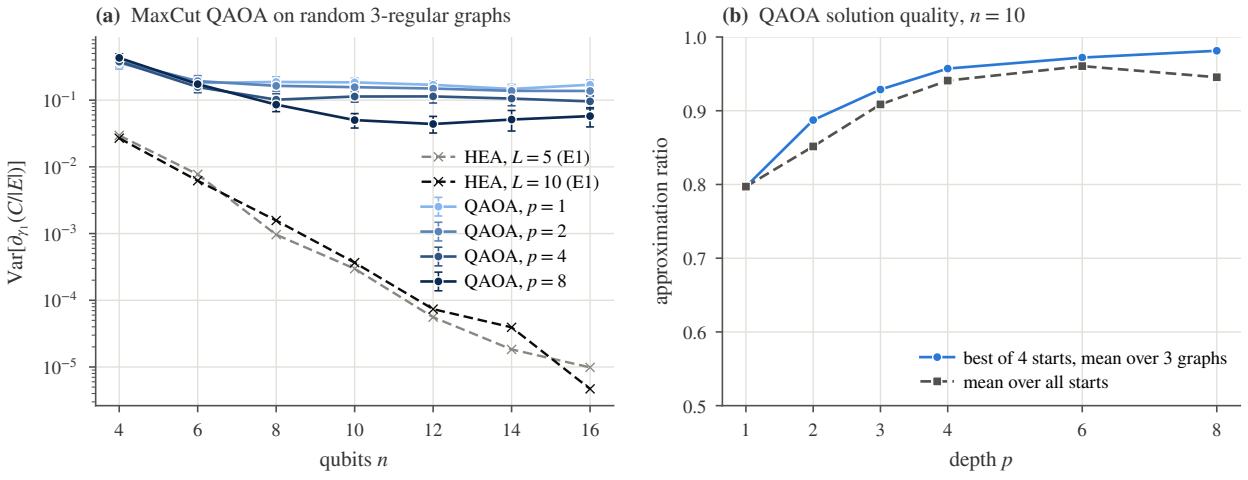
(a) MaxCut QAOA on random 3-regular graphs
Var[$\partial_{\gamma_1}\langle C/|E|\rangle$]
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
4
6
8
10
12
14
16
qubits n
HEA, $L = 5$ (E1)
HEA, $L = 10$ (E1)
QAOA, $p = 1$
QAOA, $p = 2$
QAOA, $p = 4$
QAOA, $p = 8$
(b) QAOA solution quality, $n = 10$
approximation ratio
1.0
0.9
0.8
0.7
0.6
0.5
1
2
3
4
6
8
depth p
best of 4 starts, mean over 3 graphs
mean over all starts